MedFeatureStore

A Reusable SQLite-Based Medical Image Feature Repository for Multi-Disease Classification Using Classical Machine Learning

Project Proposal

Course
CSE 445 – Machine Learning

Faculty
Dr. Mohammad Ashrafuzzaman Khan (AzK)

Section
03

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Group Members

Group Member	Student ID
Touhid Ara Himu	2312187642
Md. Shahriar Rakib Rabbi	2321937642
Tasnuva Islam Ayona	2233379642

What Problem Are We Solving?

Current Challenges

- Feature extraction is repeated every time a model is trained.
- Computational cost increases.
- Feature reuse is not possible.
- Experiments become difficult to reproduce.
- Most classical ML systems focus on only one dataset.

Our Proposed Solution

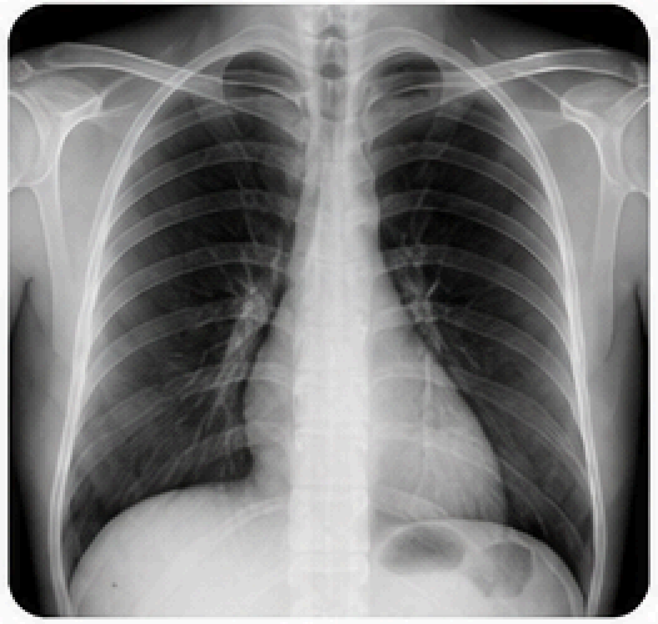
MedFeatureStore

A reusable SQLite-based repository that stores handcrafted image features only once and allows multiple machine learning models to reuse them efficiently.

Benefits

- Faster model training
- Reduced computational cost
- Reusable features
- Better reproducibility
- Modular architecture
- Multi-disease support

Supported Datasets



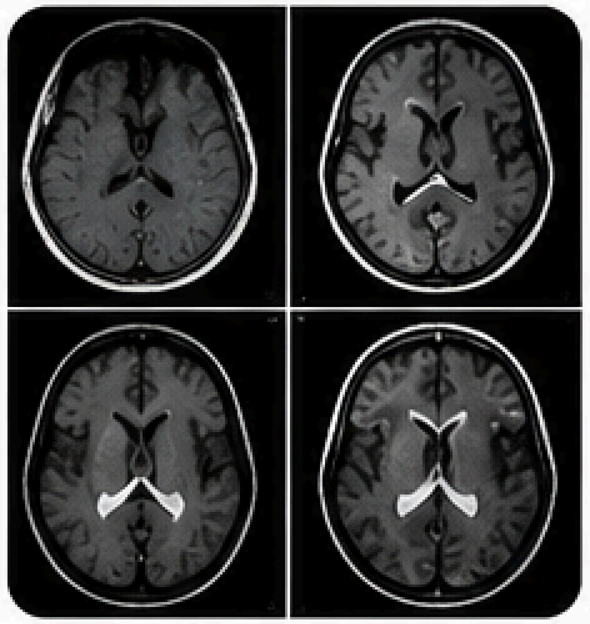
Dataset 1

Chest X-ray Pneumonia Dataset

Disease: Pneumonia

Classes:
- Normal
- Pneumonia

Dataset Link:
https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia



Dataset 2

Brain Tumor MRI Dataset

Disease: Brain Tumor

Classes:
- Glioma
- Meningioma
- Pituitary Tumor
- No Tumor

Dataset Link:
https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

Proposed Methodology

Our Approach, Methods, and Tools

1. **Medical Image Dataset**
   Collect and organize medical images from multiple sources.
2. **Image Validation**
   Remove corrupted, duplicate, irrelevant images.
3. **Image Preprocessing**
   - Resize
   - Grayscale
   - Normalization
4. **Feature Extraction**
   - HOG
   - LBP
   - GLCM
   - Color Histogram
5. **SQLite Feature Repository**
   Store extracted features in a reusable SQLite database.
6. **Feature Selection**
   Select the most informative features for model training.
7. **Machine Learning Models**
   - Logistic Regression
   - KNN
   - Decision Tree
   - Random Forest
   - Support Vector Machine
8. **Performance Evaluation**
   Evaluate models using multiple metrics and validation techniques.
9. **Explainable AI (SHAP)**
   Interpret model predictions using SHAP values.
10. **Prediction Dashboard (Streamlit)**
    Deploy the best model with an interactive web dashboard.

Technologies Used

- **Google Colab** — Cloud-based development environment
- **Python** — Programming language
- **scikit-image** — Image processing tools
- **scikit-learn** — Machine learning library
- **SQLite** — Lightweight database
- **SQLAlchemy** — ORM for database interaction
- **SHAP** — Model explainability library
- **Streamlit** — Interactive web app framework
- **GitHub** — Version control & collaboration

The proposed architecture extracts handcrafted image features only once, stores them in a reusable SQLite repository, and enables multiple classical machine learning models to reuse the same features. This reduces redundant computation while improving reproducibility and training efficiency.

Evaluation Strategy

How will project success be measured?

Machine Learning Evaluation

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

These evaluation metrics measure the classification performance of each machine learning model.

Repository Performance Comparison

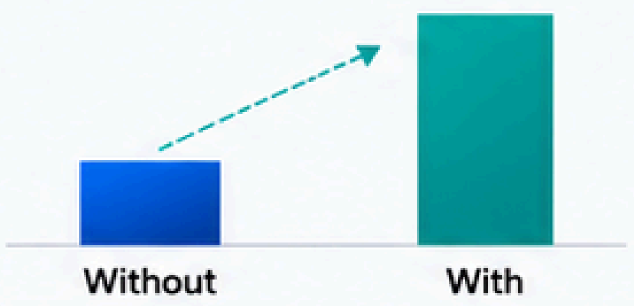
Without MedFeatureStore	With MedFeatureStore
Repeated Feature Extraction	Reusable Features
Longer Training Time	Faster Feature Loading
Higher Computational Cost	Lower Computational Cost
Difficult Experiment Reproducibility	Better Reproducibility
Duplicate Feature Generation	One-Time Feature Storage

Efficiency Improvement

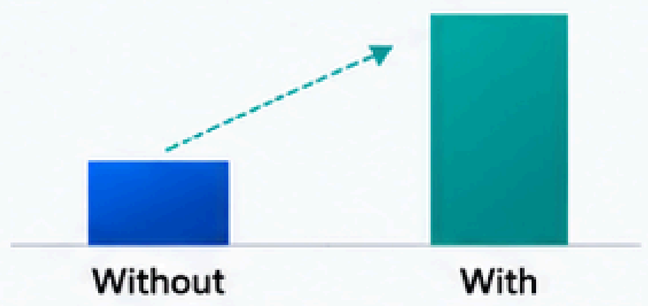
Training Time
(↓ Lower is Better)

Computational Cost
(↓ Lower is Better)

Feature Reusability
(↑ Higher is Better)



Experiment Reproducibility
(↑ Higher is Better)

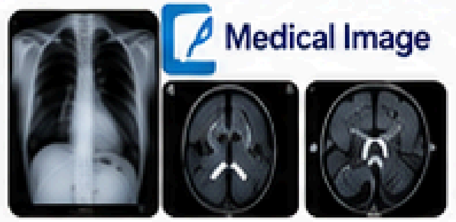


Goal: Develop an efficient, reusable, and reproducible classical machine learning pipeline for multi-disease medical image classification.

Project Deliverables

Expected Outputs and Contributions

- Google Colab Notebook
- SQLite Feature Repository
- Medical Image



- Extracted Feature Database
- Trained Machine Learning Models
- Performance Evaluation Report
- SHAP Explainability Analysis
- Streamlit Prediction Dashboard
- GitHub Repository
- Final Project Report (20–30 Pages)
- Presentation Slides
- Live Project Demonstration

Expected Contributions & Milestones

- Reusable Feature Repository
- Reduced Feature Extraction Time
- Improved Experiment Reproducibility
- Multi-Disease Classification
- Modular Machine Learning Framework
- Explainable AI Integration
- Easy Extension to New Datasets and Models

★ **MedFeatureStore** aims to provide a reusable, efficient, and scalable feature management framework that enhances classical machine learning workflows for medical image classification while improving computational efficiency, reproducibility, and model transparency.